
Oliver Krauss

Convolutional Neural Networks

AIN2IL | MTD.ba SS2026

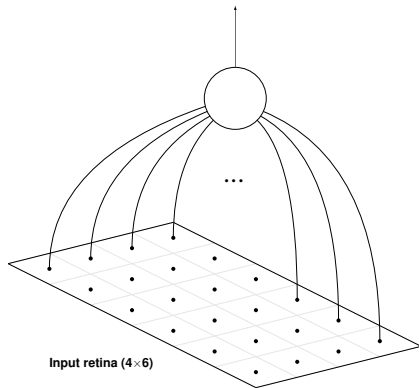
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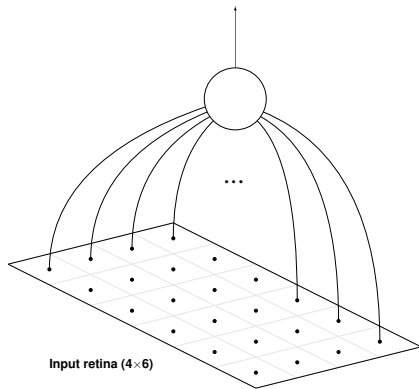


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ANNs for Image Analysis?



ANNs for Image Analysis?



How would that work for large images?

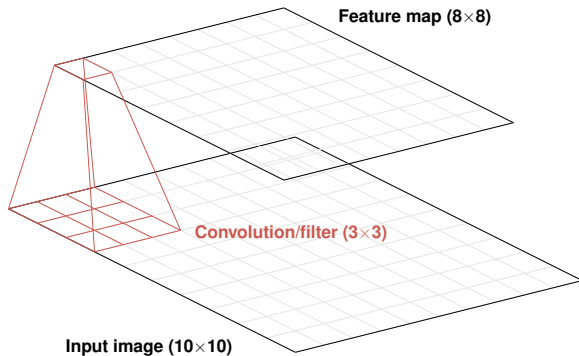
CNNs: Motivation

- In principle, classical feed-forward neural networks (fully connected networks) could be used for image analysis by simply connecting the pixels to input units.
- However, if at all, this only makes sense for small aligned images (e.g. in character recognition).
- For the analysis of larger and more complex images, this standard architecture is not useful.
- Instead, it is common to have stacked layers of units that operate on small overlapping patches/windows. Such networks are called *convolutional neural networks (CNNs)*.

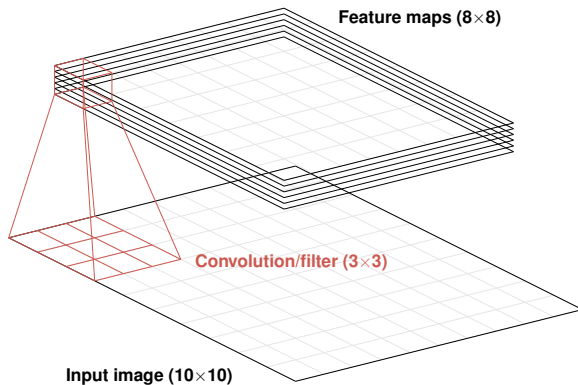
CNNs: Architecture

- The first convolutional layer usually consists of multiple units that operate on small image patches (3×3 , 5×5 , or 7×7).
- Each unit corresponds to one simple feature of a patch. Such units are often called *filters*.
- The activations of all units are computed for all patches, thereby creating a *feature map* of the image.
- Convolutional layers can be stacked.

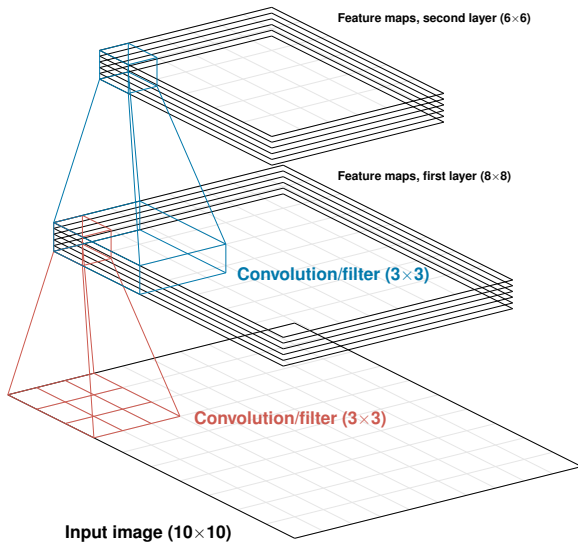
CNNs: Architecture



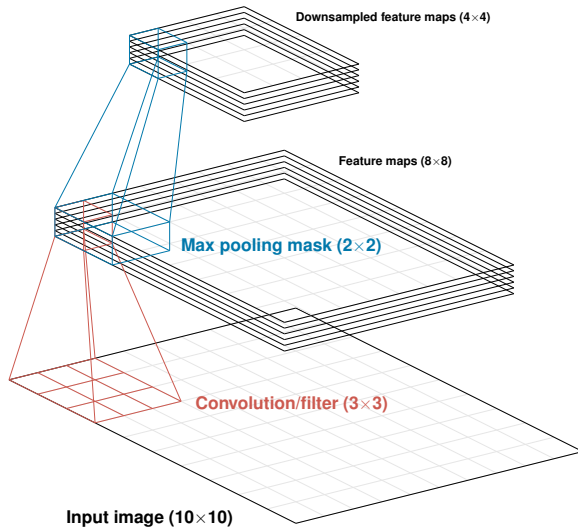
CNNs: Architecture



CNNs: Architecture



CNNs: Architecture



CNNs: Additional Layers

Batch normalization: To increase the stability of a neural network and to speed up learning, batch normalization normalizes the activations of a previous layer to mean 0 and standard deviation 1.

Pooling: It can be useful to down-sample feature maps by local *max pooling* (e.g. with non-overlapping 2×2 windows).

Standard Output vs. Fully Convolutional

- Fully connected (dense) layers:** Networks that perform traditional classification (or regression) with image inputs need a fixed number of outputs. To this end, it is common to *flatten* the last convolutional layer and to connect it to the output units like in a standard multi-layer perceptron, possibly with some hidden layers in between.
- Fully convolutional network:** Networks that also yield images as outputs may also consist of convolutional layers only (e.g. for (semantic) image segmentation).

CNNs: Training

- Fully connected layers, if any, are trained as usual.
- In convolutional layers, each feature map/filter has only one set of weights and all windows contribute weight updates. This is called *weight sharing*.
- In max pooling layers, the error signal is only propagated to the input from which the maximal activation came.

How to Learn Good Representations?

- The success of a deep network is determined by how meaningful the representations in the hidden layers are.
- What is a meaningful representation?
 - Each convolutional unit/filter corresponds to a specific pattern (hidden) in the data.
 - Each convolutional layer corresponds to a certain *level of abstraction*.

Layered Representations in a CCN: Example

Two layers of a convolutional network: hypothetical inputs maximizing activation and real images that lead to a high activation of the considered neuron

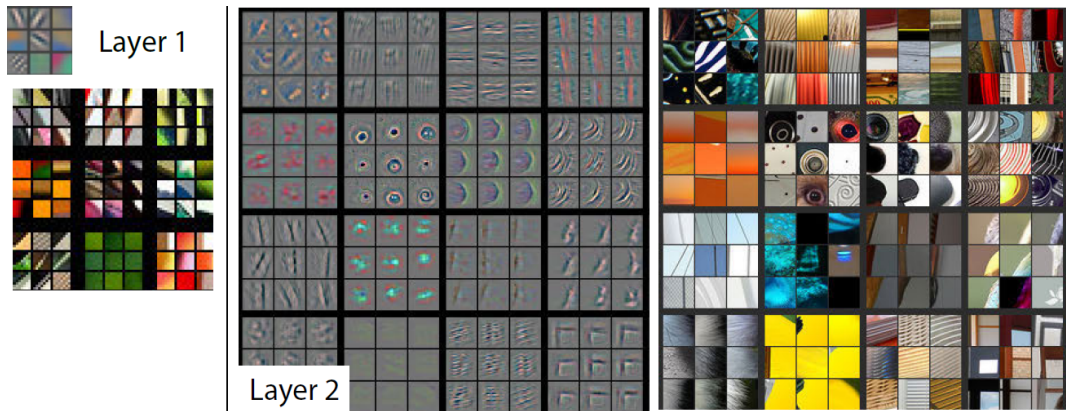


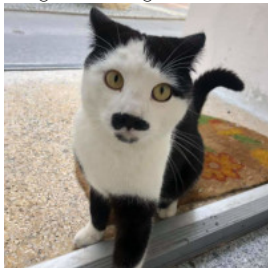
Figure: Source: M.D. Zeiler & R. Fergus, arXiv, 2013

Data Augmentation

- In many image classification tasks, the presence of an object is invariant to some transformations of the image. As an example, a cat still looks like a cat if the image is slightly shifted, rotated, or horizontally flipped.
- If training images are transformed before feeding them into the network, the network learns to recognize objects also if they are slightly shifted, rotated, etc.
- This often leads to better generalization properties of the networks. In some sense, this is a form of *regularization by training with noise*.
- The deliberate random transformation of input data is often called *data augmentation*.

Data Augmentation: Example

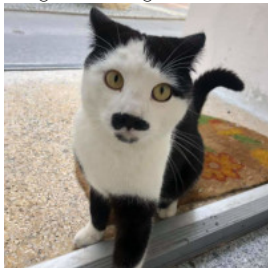
Original image:



Data Augmentation: Example

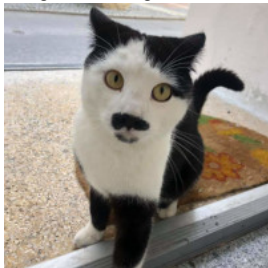
Augmented images:

Original image:



Data Augmentation: Example

Original image:



Each time an image is used in a training batch, a different transformation is applied.

Augmented images:



ImageNet

- ImageNet is a large annotated image database designed for use in computer vision research and development.
- The database consists of about 14 million images annotated with more than 20,000 categories of objects (whether or not an image contains an object of that category). About 1 million images are also annotated with bounding boxes (if an image contains an object of that category, in which surrounding rectangle the object occurs).
- ImageNet is used in the *ImageNet Large Scale Visual Recognition Challenge (ILSVRC)*.



ILSVRC Error Rates

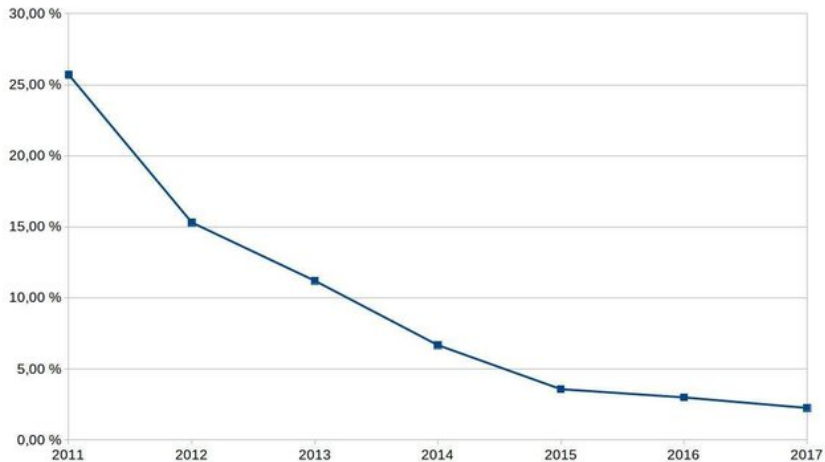


Figure: Source: European Commission, Report number EUR 29442, 2018

ILSVRC Error Rates

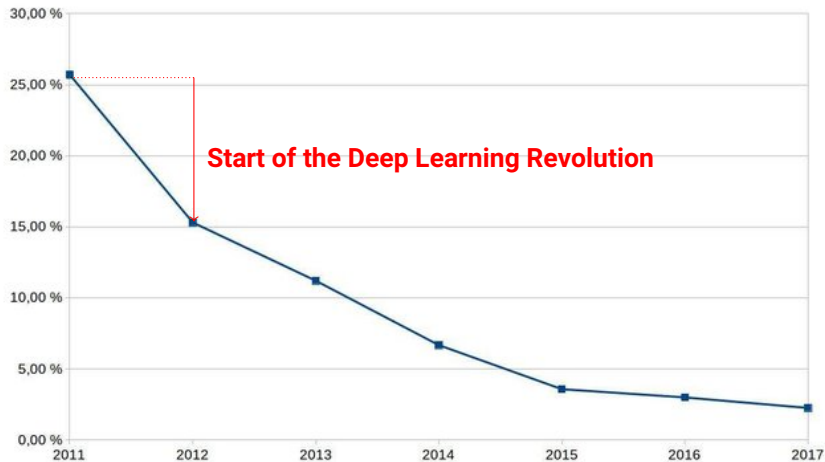


Figure: Source: European Commission, Report number EUR 29442, 2018

ILSVRC Error Rates

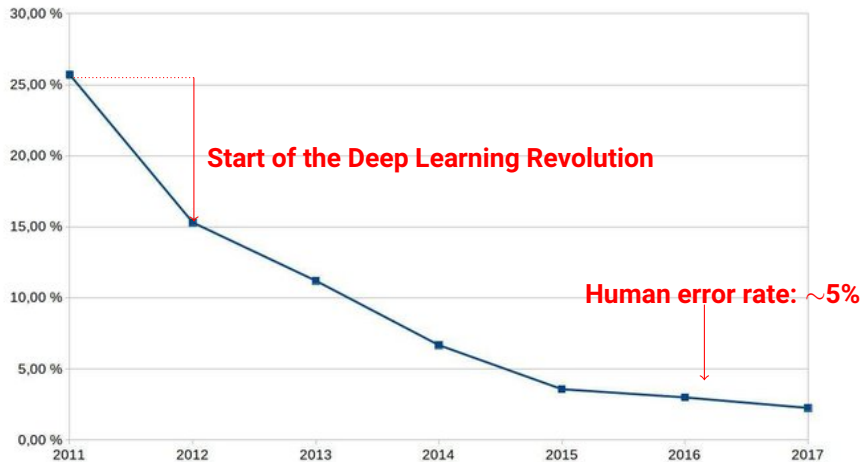


Figure: Source: European Commission, Report number EUR 29442, 2018

ILSVRC Error Rates

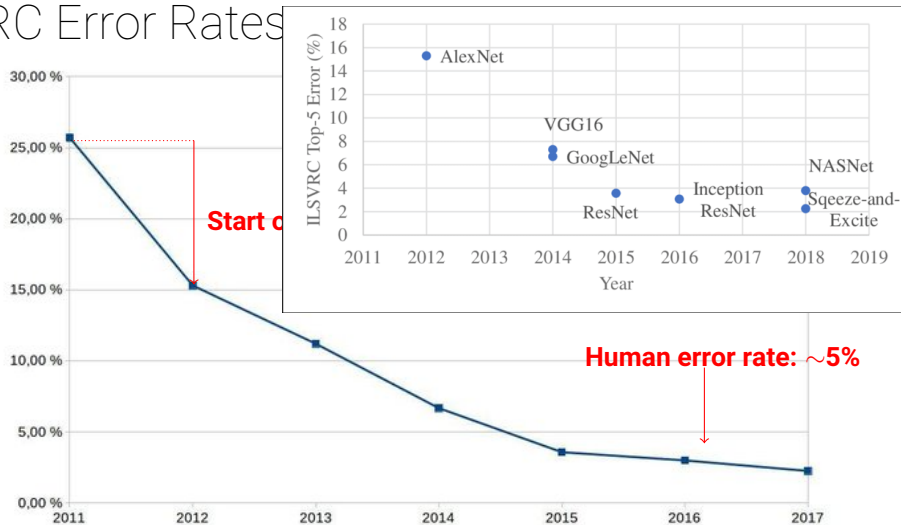


Figure: Source: European Commission, Report number EUR 29442, 2018

AlexNet Architecture

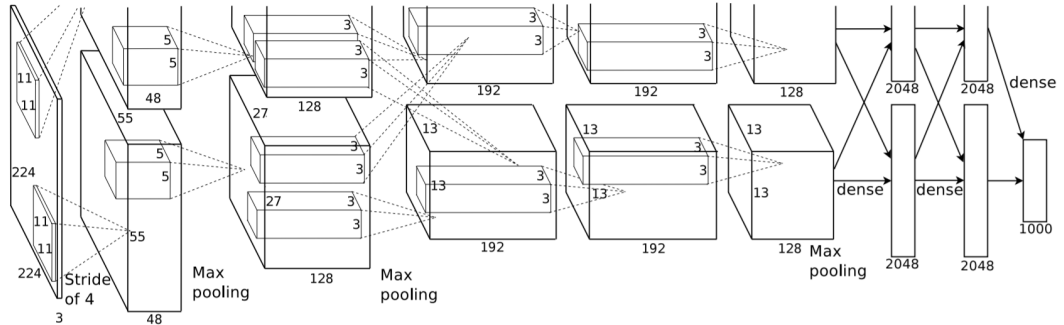


Figure: Source: A. Krizhevsky, I. Sutskever & G.E. Hinton, NIPS 2012

Other Successful Deep CNNs

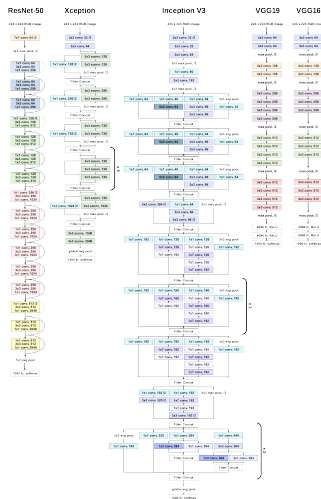
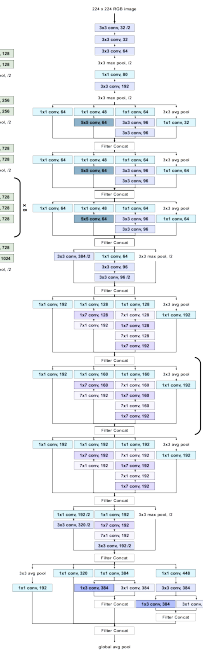
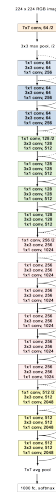


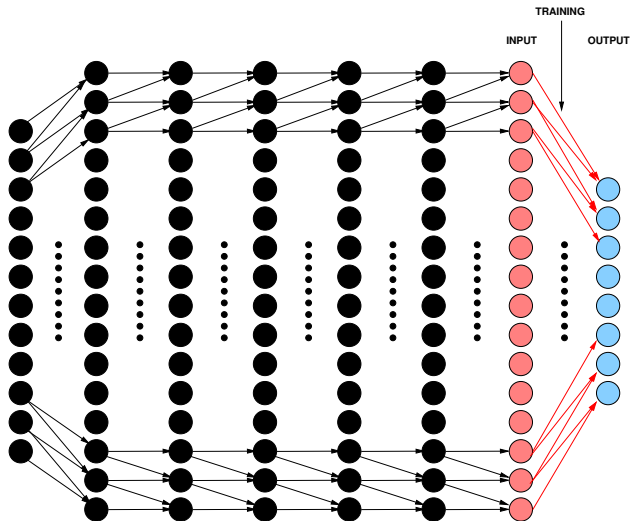
Figure: Source: T. Carvalho et al., SIBGRAPI 2018



Using Pre-Trained Networks I

- The networks mentioned previously have been trained on the massive ImageNet data set and have been engineered to create sensible layered representations of almost any type of abstract features in image data.
- These representations can be useful for any other image classification task in which similar features are present --- even if significantly less training images are available for this task.
- It has become a common and highly successful approach to re-use pre-trained networks for new task:
 1. Chop off output layer or last ***n*** layers.
 2. Add new layers on top.
 3. Train weights of new layers to solve new tasks. The pre-trained weights can be fixed or also re-trained for the new tasks.

Using Pre-Trained Networks II



Skin Cancer Classification with CNNs

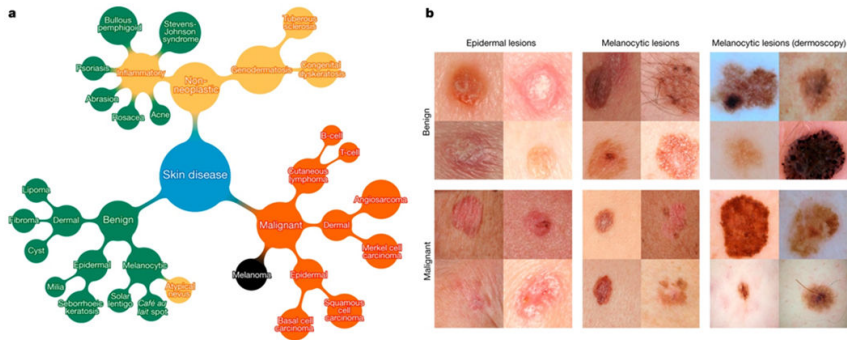


Figure: Source: A. Esteva *et al.*, Nature, 2017

A CNN based on a pre-trained Inception v3 network that was fine-tuned on approx. 129,400 training images showing skin lesions achieved superhuman accuracies (Univ. Stanford).

Object Detection With CNNs

- So far, we have mainly considered CNNs with standard output layers, i.e. each image is assigned to one (or several) classes. This approach can be used to classify images or image patches, but does not allow for localizing objects.
- However, with some modifications, object detection with localization can be achieved also with CNNs.

Recognition Based CNNs (R-CNN)

- General idea: CNNs are used on image patches, so-called *regions of interest (Rols)*. A binary SVM (one per class) is used to compute meaningful Rols.
- The CNN is pre-trained on ImageNet bounding boxes, where an additional class corresponding to “no object of interest” is added.
- Typically, a large number of Rols is used (~ 2000).
- Box selection and bounding box regression are employed as post-processing steps.
- R-CNN is quite slow.

R-CNN: Simplified Workflow

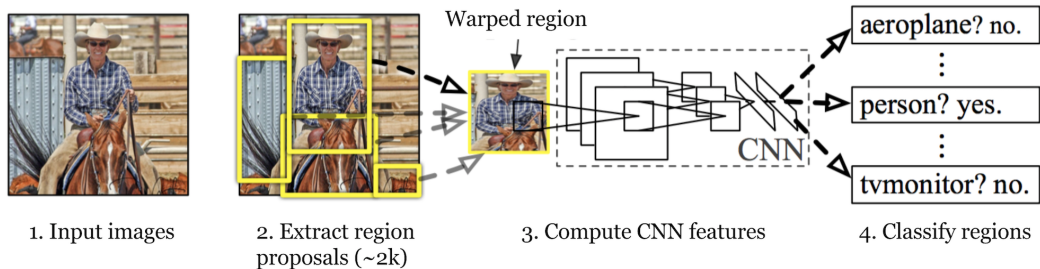
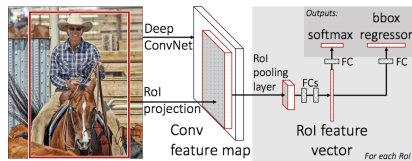


Figure: Source: R. Girshik *et al.*, arXiv, 2014

R-CNN: Variants

Fast R-CNN: Instead of SVMs, Fast R-CNN uses a CNN to generate a feature map from which Rols are extracted.



Faster R-CNN: Uses an integrated CNN for generating region proposal.

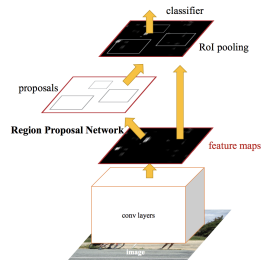


Figure: Source: R. Girshik *et al.*, arXiv, 2014

You Only Look Once (YOLO)

- The input image is divided into $S \times S$ patches (e.g. 7×7).
- For each patch, a certain number of bounding boxes with confidence scores and class probabilities is computed.
- Of all the bounding boxes, only those with high confidence scores are kept; additionally, some non-maximum suppression is applied.
- **YOLOv2**: faster, but weaker detection performance
- **YOLOv3**: uses multi-task instead of multi-class (i.e. classes are not mutually exclusive); as fast as YOLOv2, but much better detection performance.

YOLO Architecture

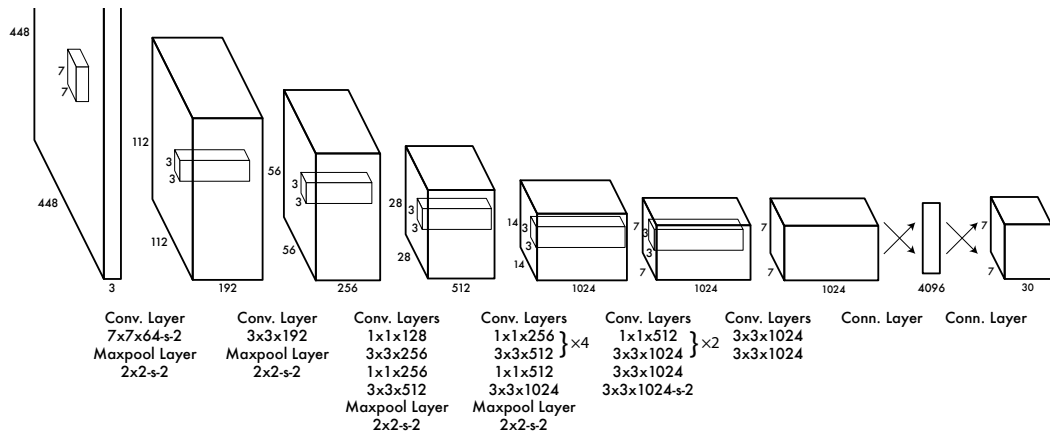


Figure: Source: J. Redmond *et al.*, arXiv, 2016

YOLO Architecture

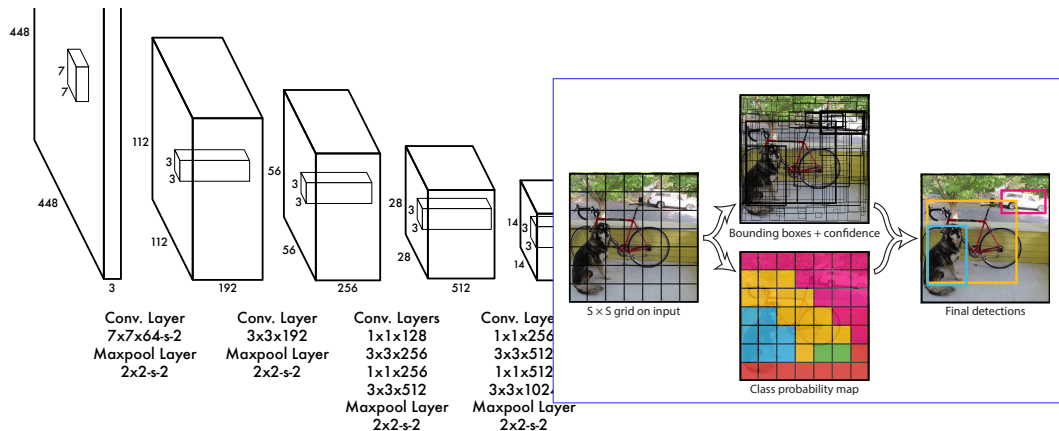


Figure: Source: J. Redmond *et al.*, arXiv, 2016

YOLO: Examples

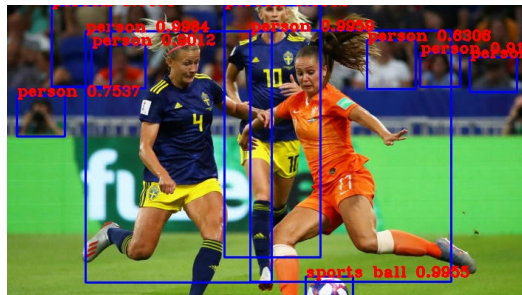


Figure: Sources (of original images): (1) Matthias Balk/DPA/Pixsell | (2) Reuters

Example: Semantic Segmentation



U-NET: Fully Convolutional Architecture for Semantic Segmentation

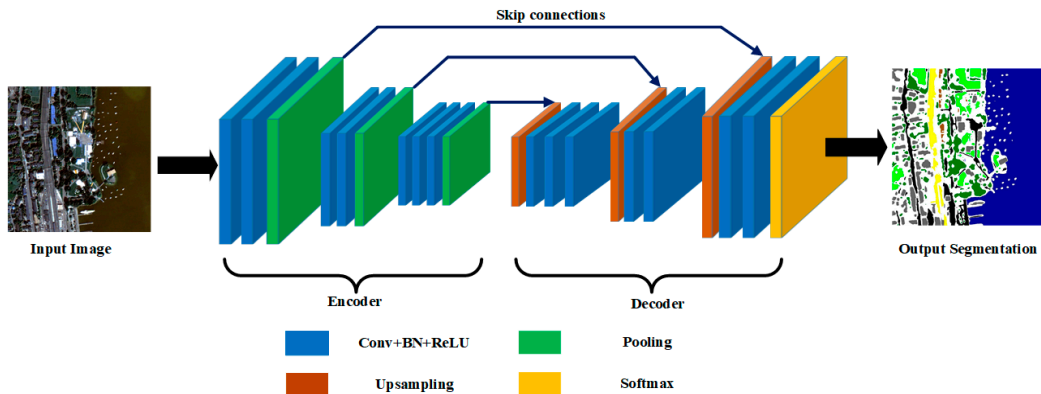


Figure: Source: W. Li et al., Remote Sensing, 2019

Semantic Segmentation of Retinal Blood Vessels

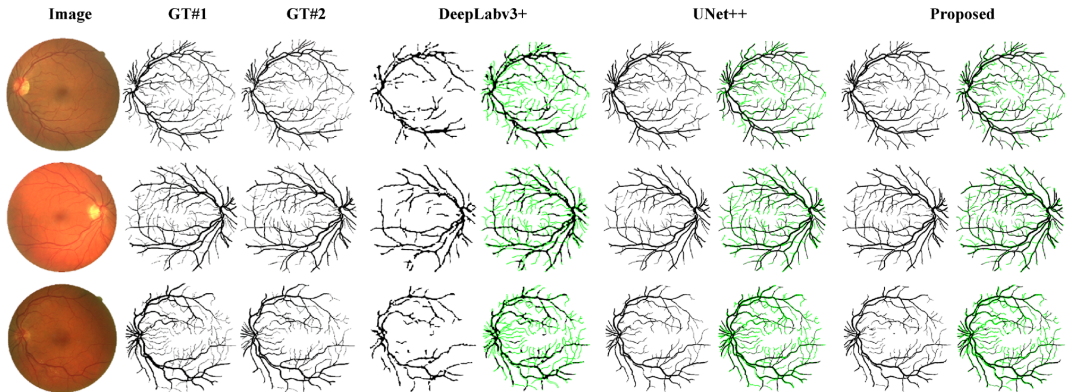


Figure: Source: Al-masni & Kim, Scientific Reports, 2021

CNNs: Final Words

- CNNs have become standard in visual object recognition --- completely disrupting the computer vision domain (ILSVRC has been dominated by CNNs since 2012).
- Deep learning is an essential ingredient in recent attempts to autonomous driving, e.g. for semantic segmentation and even end-to-end learning of driving decisions (see NVIDIA demo video).
- CNNs are even used for audio processing by applying them to so-called spectrograms.
- Combinations with recurrent neural networks and generative adversarial networks provide unprecedented possibilities.